Taejoon Han

Master's Student, Department of Computer Science and Engineering
Ulsan National Institute of Science and Technology (UNIST)
Email: taejoon.han@unist.ac.kr

RESEARCH INTERESTS

Graph mining and data management, with a focus on cohesive subgraph discovery, community search, and network stability analysis on large-scale and complex networks.

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) Sep 2025 – Present
M.S. in Computer Science and Engineering
Advisor: Prof. Junghoon Kim

Ulsan National Institute of Science and Technology (UNIST) Mar 2020 – Aug 2025
B.S. in Computer Science and Engineering

AWARDS & HONORS

- **Research Subsidies for Master's Students, NRF** 2026 Fall – 2027 Summer
- **ACM SIGMOD 2026 Travel Grant Awardee** 2026
- **ACM CIKM 2025 Student Travel Grant**. 2025
- **National Scholarship for Graduate Studies**, Ministry of Education, South Korea. 2025 Fall – Present
- **Best Paper Award (Bronze)**, Korea Database Conference (KDBC) 2024. 2024

PUBLICATIONS

Underlined and bold *indicates the author of this CV.* * *denotes equal contribution.*

International Conferences & Journals

- Minseok kim, Dahee Kim, **Taejoon Han**, Junghoon Kim, Seungil Wi, and Jonghyeok Park. "The s-Hypercore: A Cohesive Subhypergraph Model Based on Decayed Engagement Strength" *IEEE TKDE 2027* (Top database journal, SCIE Q1).
- **Taejoon Han**, Dahee Kim, Junghoon Kim, and Taeho Ha. "Representation-Aware Modularity with a Distance-Calibrated Null Model" *ACM CIKM 2026* (CS top conference, oral presentation).
- Dahee Kim, **Taejoon Han**, Kaiyu Feng, Junghoon Kim, and Susik Yoon. "LMSC: Local Sketch Modularity Optimisation for Size-Constrained Community Search in Networks" *ACM SIGMOD 2026* (Top database conference).
- Song Kim, Dahee Kim, **Taejoon Han**, Junghoon Kim, Hyun Ji Jeong, and Jungeun Kim. "Efficient Locality-based Indexing for Cohesive Subgraphs Discovery in Hypergraph" *EDBT 2026* (Top database conference).
- **Taejoon Han***, Song Kim*, Woungjae Choo, and Junghoon Kim. "From Bicliques to Biflexi Cliques: A New Era of Bipartite Subgraph Discovery" *ACM CIKM 2025* (CS top conference).

Domestic Conferences & Journals

- Juhyeok Seo*, **Taejoon Han***, Seungjoo Jung, and Junghoon Kim. "Minimising Edge Additions for Joint k-Core Activation of Query Nodes" *KDBC 2026*.
- **Taejoon Han** and Junghoon kim. "Towards Certified Semantic Community Search in Networks" *KDBC 2026* (Work-in-Progress track).
- **Taejoon Han**, Seungchan Choi, Minseok Kim, Hyewon Kim, Dahee Kim, and Junghoon Kim. "NHSCAN: Node-based Hypergraph SCAN" *KDBC 2025* (Work-in-Progress track).

- Seungchan Choi*, **Taejoon Han***, Dahee Kim, Hyewon Kim, Minseok Kim, and Junghoon Kim. "Weighted Structural Similarity for Density-Based Graph Clustering" *KDBC 2025* (Work-in-Progress track).
- Deokhyun Kim*, **Taejoon Han***, and Junghoon Kim. "Community Detection in Large-scale Attributed Networks via Joint Structural and Attribute Modularity Maximization" *Journal of Database Research*, 2025.
- Dahee Kim*, Hyewon Kim*, **Taejoon Han**, and Junghoon Kim. "Community Detection Based on Ollivier-Ricci Curvature and Modularity" *KDBC 2024*. **(Best Paper Award)**
- **Taejoon Han***, Hyewon Kim*, Seongsik Hwang, Woocheol Shin, Junghoon Kim, Jaehyun Bhang, and Hyeongi Hong. "Complex Network Stability: Exploring Comprehensive Anchored Edge s-Core" *KDBC 2024* (Work-in-Progress track).

PATENTS

- "Method and Apparatus for Discovering Cohesive Subhypergraphs Using Decay-based Participation Strength in Hypergraphs." Application No. 10-2026-0096258, May 2026.
- "Network Clustering System and Method Based on Weighted Structural Similarity." Application No. 10-2026-0004498, Jan 2026.
- "Community Detection Method for Large-scale Attributed Networks via Joint Structural and Attribute Modularity Maximization." Application No. 10-2025-0205480, Dec 2025.
- "Method and Apparatus for Discovering Cohesive Subgraphs in Bipartite Networks." Application No. 10-2025-0166255, Dec 2025.
- "Method and Apparatus for Community Detection Based on Ollivier-Ricci Curvature and Modularity." Application No. 10-2025-0034070, Mar 2025.

TEACHING EXPERIENCE

Teaching Assistant, CSE321 Database Systems Spring 2026
Department of Computer Science and Engineering, UNIST
Teaching Assistant, NOVA503 AI Toolkit for Industry Spring 2026
Department of Computer Science and Engineering, UNIST
Teaching Assistant, CSE221 Data Structures Fall 2025
Department of Computer Science and Engineering, UNIST
Lecturer/TA, LG Electronics DX & Living DX Intensive Course Jun 2024 – Feb 2026
Conducted multiple industry training programs for LG Electronics employees, covering Python, data mining, machine learning, and applied LLM topics. Selected sessions:

- *Data Mining & Machine Learning* (Jun 2024, Jul 2025): Lectured on Apriori, Decision Trees, Random Forests, k-Means, and DBSCAN with hands-on Python/Google Colab labs.
- *PBL — Personalized Washing Machine Course Recommendation* (Feb 2025): Mentored a project applying multi-output Random Forests and clustering for personalization based on user-level usage data.
- *PBL — Factory Power Consumption Forecasting* (Aug – Sep 2025): Guided data preprocessing, Ridge regression with lag/rolling-mean features, and built a RAG-based chatbot using OpenAI API, ChromaDB, and Streamlit.
- *Prompt Engineering with Open-source LLMs* (Jan 2026): Deployed `gpt-oss-20b` on a lab server with a custom web API and led prompt-engineering workshops covering parameter tuning, sentiment analysis, and translation.
- *BOM Price-change Automation with LLMs* (Feb 2026): Developed a tree-based BOM-comparison pipeline and an LLM-based PowerPoint specification reader for automated extraction of part-cost change reasons.

Lecturer/TA, Nuclear Power & Energy Specialized AI Professional Training Program May 2026
Conducted a hands-on session covering the full pipeline from data preprocessing with pandas to the implementation of Decision Trees, k-Means, and DBSCAN.
Lecturer/TA, Research Paper Writing — UNIST Novatus Graduate School July 2026

TA, Gyeongnam AI Novatus Academia July 2026 – October 2026
Conducted a PBL session using locally deployed Qwen models to merge tables from specification PDFs,

generate rule-based product-code combinations, and validate outputs with a second LLM.

SERVICE & ACTIVITIES

- **CSE Supporters**, UNIST. Mar 2024 – Jun 2025
- **Student Council Member**, UNIST. Sep 2020 – Jun 2022